

Freq 2 (2024)

- ① We are testing if V_2 is larger ($H_1: V_2 > V_1$), so this is a right tailed test. We want a positive test statistic



z is here, so the area to the left of z is 0,0094 (according to the table). Which means that the area to the right is $1 - 0,0094 = 0,9906$.

$p\text{-value} = 0,9906$ (way larger than any standard α , so we retain H_0)

- ② Ex 15 from the practice exercises.

②

1- The t-distribution is normal. If the data is skewed, the sample means will also be skewed ($n < 30$ so the Central Limit Theorem does not take effect). Feeding this asymmetrical data breaks the model.

2- When we look up t_c in the t-table to find the p-value, we are looking at the AUC for a normal distribution. If the sample was actually drawn from another non-normal distribution, the true curve is different and the p-value is wrong.

- Type I errors increase if, for example, the table tells us that $p = 0,03$, but because our true data is more concentrated at the tails, the real p-value might be $p = 0,12$ (we incorrectly reject H_0)

- Type II errors increase if the reverse occurs (table says $p = 0,12$ but our curve is narrower at the tails and the real p-value is 0,03)

③ $H_0: \mu_d \leq 0$ $d = V1 - V2$

④ $H_1: \mu_d > 0$ $\alpha = 0,05$

Ignore missing value ($n=8$)

$d = \{ 4.9, -2.2, 1.2, 5.2, -1.6, 2.2, 1.1, 1.6 \}$

$d^2 = \{ 3.61, 10.24, 1.44, 22.04, 2.56, 4.84, 1.21, 2.56 \}$

$\sum d_i = 8,9$ $\bar{d} = \frac{\sum d_i}{n} = 1,05$

$\sum d_i^2 = 53,5$ $s_d = \sqrt{\frac{n(\sum d_i^2) - (\sum d_i)^2}{n(n-1)}} \approx 2,526$

$t_c = \frac{\bar{d} - \mu_0}{\left(\frac{s_d}{\sqrt{n}}\right)} = \frac{1,05}{0,873} \approx 1,176$

$df = 8 - 1 = 7$ $0,1 < p\text{-value} < 0,15$

$p\text{-value} > \alpha \Rightarrow$ Retain H_0

- ⑤ Yes, it's better. Eliminates
- Eliminates Learning. For example, if a participant got better with V2 be they learned from V1
 - Eliminates bias from expectation. Participants might be positively biased if they know that they are using an "improved system"
 - Fair baseline (no previous experience)

④ $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$
 H_1 : At least one is different ($\alpha = 0,05$)

(Again, in my opinion, these are dependent samples, but we will assume that they are independent).

$$n = 8 + 8 + 8 + 8 = 32$$

$$GM = \frac{\bar{x}_1 n_1 + \bar{x}_2 n_2 + \bar{x}_3 n_3 + \bar{x}_4 n_4}{32} = 6,1175$$

$$SS_b = \sum n_i (\bar{x}_i - GM)^2 = 10,9376$$

$$SS_w = \sum (n_i - 1) s_i^2 = 34,0711$$

$$df_b = k - 1 = 3$$

$$df_w = n - k = 28$$

$$MS_b \approx 3,6459$$

$$MS_w \approx 1,2168$$

$$F = \frac{MS_b}{MS_w} \approx 2,996$$

$$F^*(\alpha, df_b, df_w) = 2,9467$$

$$F^* < F\text{-value} \Rightarrow \text{Reject } H_0$$